6 **(a)** Magnetic field due to current in wire B is normal to the current in wire A, and pointing into plane of paper. By Fleming's left hand rule, this causes a magnetic force to be exerted on wire A towards wire B. B1

Based on Newton's 3rd law, a magnetic force is also exerted on wire B by wire A which is of the same magnitude but opposite in direction, giving rise to an attractive force between both wires. B1

(b) (i) $B = \dfrac{\mu_0 I_A}{2\pi d}$

$= \dfrac{4\pi \times 10^{-7}(90)}{2\pi(0.50)}$

$= 3.6 \times 10^{-5}$ T M1

$Bqv \sin 90^0 = \dfrac{mv^2}{r}$ M1

$r = \dfrac{mv}{Bq} = \dfrac{(1.67\times10^{-27})(1.0\times10^{3})}{(3.6\times10^{-5})(1.6\times10^{-19})}$

$= 0.29$ m A1

(ii) As proton is nearer to wire A, B increases ($B \alpha \dfrac{1}{r}$**)** and radius decreases due to the increasing magnetic force ($F = Bqv \sin\theta$). B1

Eventually at the nearest location to wire, the velocity of proton is parallel to wire, therefore force is directed away from wire, radius is smallest and proton is turned back. B1

B decreases further from wire A and radius increase due to decreasing *F*. B1

7 (a) (i) EM produced whenever charged particle is suddenly accelerated/ decelerated at the